Structural and Semantic Decomposition for Robust News Article Classification

Matteo Sisti
Politecnico di Torino
Student id: s362355
matteo.sisti@studenti.polito.it

Beybin Çiçek
Politecnico di Torino
Student id: s351818
beybin.cicek@studenti.polito.it

***Abstract*—This project addresses a multi-class news article classification task on a real-world, HTML-heavy dataset with noisy markup, strong editorial bias, and intrinsic label ambiguity, evaluated using macro-averaged F1 score.**

Exploratory analysis motivates a Two-Stage strategy that separates structurally deterministic articles from semantically ambiguous ones, highlighting the limits of linear models.

RoBERTa and DeBERTa transformers are fine-tuned and combined through decision-level ensembles, where majority voting is used to reinforce robustness rather than peak performance, and selective routing improves predictions on hard samples.

I. Problem Overview

This project addresses a multi-class news article classification task, where each document is assigned to one of seven topical categories. The dataset consists of real-world online news articles in raw HTML format, characterized by heterogeneous sources, noisy markup, incomplete metadata, and strong editorial regularities [1]. Performance is evaluated using macro-averaged F1 score to ensure balanced behavior across classes [2].

A preliminary exploratory analysis is conducted to identify informative, redundant, and structurally biased signals, guiding preprocessing and modeling choices.

a) Temporal Metadata and Redundancy: The *timestamp* attribute contains a high proportion of missing values (**34.7%**) and exhibits coarse temporal granularity, with no clear relationship between fine-grained temporal components and the target labels (see Fig. 1). An auxiliary experiment shows that publication year can be predicted directly from raw article text with approximately **64.0%** accuracy, indicating that temporal information is implicitly encoded in lexical and editorial patterns. Consequently, explicit timestamp information is treated as redundant and excluded.

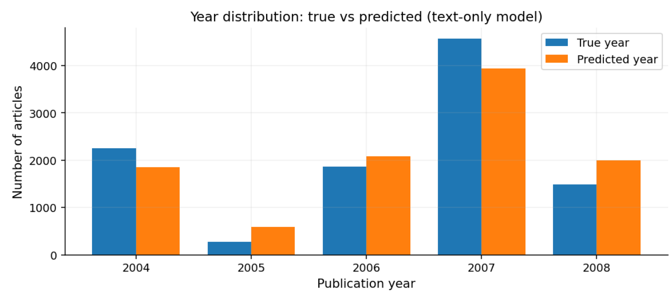

Fig. 1: Distribution of publication years in the development set and corresponding predictions obtained from a text-only classifier, confirming that temporal information is implicitly encoded in the text.

b) Non-informative Metadata: The *page_rank* feature exhibits a highly concentrated distribution. Although a moderate association with the labels is observed ($\eta^2 \approx 0.29$), this effect is likely attributable to editorial or source-related biases rather than semantic information. For this reason, *page_rank* is not included in the final models.

c) Structural Ambiguity: Duplicated or republished content accounts for **6.87%** of development instances with identical (*title*, *article*) pairs. At the level of unique pairs, **1.97%** of distinct articles exhibit conflicting label assignments, introducing intrinsic label noise and an effective upper bound on performance [3]. Fig. 2 shows that duplicated content (only article) is unevenly distributed across labels, indicating class-dependent structural bias.

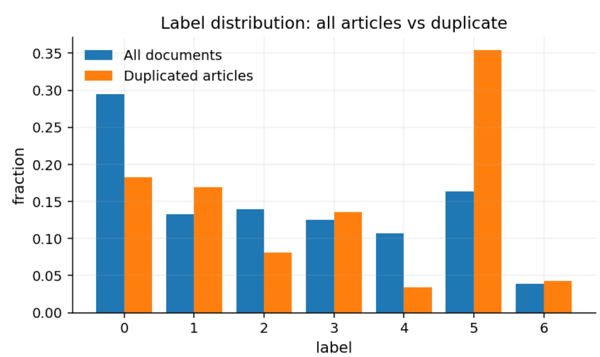

Fig. 2: Per-labels distribution in the full dataset (blue) versus duplicated articles only (orange), with duplicates defined by identical article content, normalized over duplicated instances.

d) Editorial Bias and Source Informativeness: A strong dependency exists between article source and target labels, reflecting clear editorial specialization across publishers. As shown in Fig. 3, several sources exhibit highly skewed label distributions, with some categories being almost exclusive to specific publishers. This confirms that the news source encodes strong structural and editorial signals.

One-hot encoding of the *source* attribute yields a strong linear baseline, revealing near-deterministic editorial cues and motivating modeling strategies that explicitly separate structurally deterministic cases from semantically ambiguous documents.

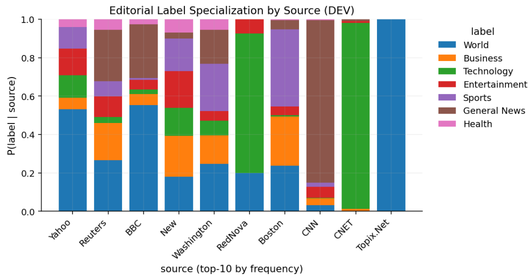

Fig. 3: Label distribution conditioned on the news source for the top-10 most frequent publishers in the development set. Each bar represents $P(\text{label} \mid \text{source})$, highlighting strong editorial specialization and source-dependent class biases.

e) Implications: Overall, the dataset combines noisy HTML-heavy text, redundant metadata, and strong editorial regularities. These properties motivate a progressive modeling approach that exploits high-confidence structural signals while reserving more expressive models for genuinely ambiguous samples.

II. Proposed approach

The proposed approach combines structural and semantic modeling strategies to address dataset heterogeneity. Editorial patterns are captured through a Two-Stage classification pipeline, while Transformer-based models model high-level semantics. Final predictions are obtained by combining complementary models to improve robustness and generalization

A. Preprocessing

Preprocessing is lightweight, using concatenated and lowercased title and article text. Linear and Two-Stage models incorporate simple numeric and source features, while Transformer-based models rely exclusively on raw text (title plus article).

B. Model selection

1) Editorial-Oriented Two-Stage Paradigm: A linear TF–IDF baseline using word- and character-level n-grams, source indicators, and simple numeric features is first evaluated [4] [5]. Despite extensive hyperparameter exploration (Table I), performance saturates around a macro-averaged F1 score of 0.719, indicating that a single-stage linear formulation largely exhausts the available linear signal.

TABLE I: Optimal hyperparameter ranges for the linear TF–IDF baseline.

Hyperparameter	Selected value / range
Word n-grams	2
Character n-grams	5–7
Regularization C	0.6–1.0
Max document freq.	0.85–0.9

Analysis of learned feature weights shows that the most influential predictors are editorial and structural cues (e.g., publisher identifiers, RSS markers, and HTML fragments), several of which display near-deterministic associations with specific classes (Table II).

TABLE II: Examples of high-support editorial tokens with near-zero conditional entropy.

Token	Type	Support	Entropy	Dominant label
<h4>	editorial	628	0.00	2
</h4>	editorial	628	0.00	2
<p align="right">	editorial	311	0.00	2
<a href=...>	editorial	233	0.00	2
<img class="rss-ad">	editorial	196	0.00	2
<strong>	editorial	111	0.00	2
<cite>	editorial	40	0.00	2
<em>	editorial	57	0.47	2

Let $\mathcal{Y}$ denote the set of class labels and $\mathcal{T}$ the token vocabulary extracted from the development set. For a token $t \in \mathcal{T}$, we estimate the empirical class distribution $p(c \mid t)$ as the fraction of development documents containing t that belong to class $c \in \mathcal{Y}$. The conditional entropy is defined as:

$$H(Y \mid t) = -\sum_{c \in \mathcal{Y}} p(c \mid t) \log p(c \mid t).$$

Tokens with $H(Y \mid t) = 0$ deterministically identify a class in the development set.

Based on this observation, we adopt a Two-Stage classification paradigm [4]. For each token t, we define:

$$\text{purity}(t) = \max_{c \in \mathcal{Y}} p(c \mid t), \qquad \text{support}(t) = \left|\{x_i \in \mathcal{D}_{\text{dev}} : t \in x_i\}\right|,$$

where $\mathcal{D}_{\text{dev}}$ denotes the development set. Tokens satisfying purity and support thresholds τ_p and τ_s are included in the rule set:

$$\mathcal{R} = \{t \in \mathcal{T} : \text{purity}(t) \geq \tau_p \ \wedge \ \text{support}(t) \geq \tau_s\}.$$

a) Stage 1: Deterministic Assignment: Given a document x, if there exists $t \in \mathcal{R}$ such that $t \in x$, the document is assigned:

$$\hat{y}^{(1)}(x) = \arg\max_{c \in \mathcal{Y}} p(c \mid t).$$

b) Stage 2: Probabilistic Classification: All remaining documents are classified by a probabilistic model $f_\theta : \mathcal{X} \to \mathbb{R}^{|\mathcal{Y}|}$:

$$\hat{y}^{(2)}(x) = \arg\max_{c \in \mathcal{Y}} f_\theta(x)_c.$$

c) Unified Decision Rule:

$$\hat{y}(x) = \begin{cases} \hat{y}^{(1)}(x), & \text{if } \exists\, t \in \mathcal{R} \text{ such that } t \in x, \\ \hat{y}^{(2)}(x), & \text{otherwise.} \end{cases}$$

This formulation explicitly separates deterministic editorial patterns from semantically ambiguous instances, while reducing to a standard single-stage classifier when no high-purity rules apply. [6]

2) Semantic-Oriented Transformer Models: Transformer-based models are used to complement the editorial bias of the Two-Stage classifier by modeling deeper semantic information. We fine-tune RoBERTa and DeBERTa using only the concatenated title and article text, without additional features [7], [8]. Predictions are treated independently and later combined through ensemble strategies to exploit semantic complementarity and improve robustness [9].

C. Hyperparameter Tuning

Hyperparameter tuning is conducted to assess the sensitivity and robustness of the proposed Two-Stage classifier, rather than to aggressively maximize performance.

For Transformer-based models, hyperparameter choices follow standard fine-tuning practices and are primarily used to verify convergence stability, rather than to perform exhaustive optimization.

1) Two-Stage Rule Parameters: We analyze the sensitivity of the Two-Stage classifier to its main hyperparameters: the minimum rule purity τ_p, the minimum rule support τ_s, and the regularization parameter C of the Stage 2 classifier.

Figure 4 summarizes the effect of these parameters on macro-averaged F1. Panel (a) shows a monotonic increase in performance as τ_p increases, confirming that selecting highly pure patterns consistently improves accuracy. Panel (b) indicates that performance remains largely stable across a broad range of τ_s, suggesting limited sensitivity once sufficiently frequent rules are retained. Finally, Panel (c) shows that macro-F1 is maximized for moderate values of C, with only marginal degradation for larger values.

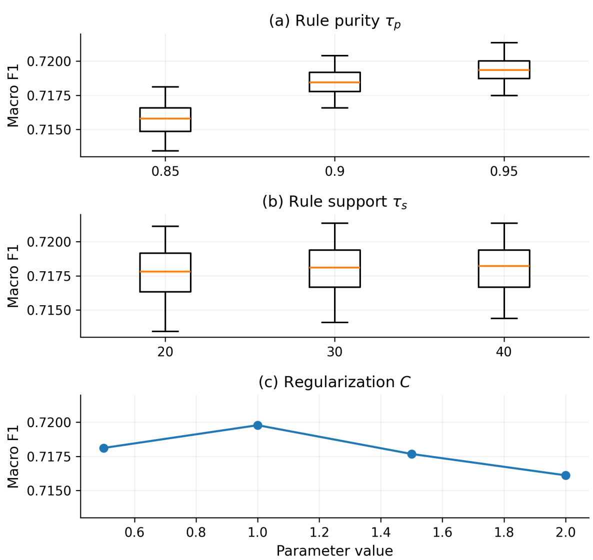

Fig. 4: Sensitivity of the Two-Stage classifier to its main hyperparameters. (a) Macro-F1 vs. rule purity threshold τ_p. (b) Macro-F1 vs. rule support threshold τ_s. (c) Macro-F1 vs. regularization parameter C of the Stage 2 classifier.

a) Best Configurations: Table III reports the top-performing configurations on the development set, together with the corresponding rule coverage.

TABLE III: Top-performing Two-Stage configurations on the development set. Reported are the regularization parameter C, minimum rule purity τ_p, minimum rule support τ_s, macro-averaged F1 score, and rule coverage.

C	τ_p	τ_s	**Macro F1**	**Coverage**
1.0	0.95	40	0.7214	0.156
1.0	0.95	30	0.7214	0.162
1.0	0.95	20	0.7211	0.172
1.0	0.90	30	0.7204	0.189
1.0	0.90	20	0.7202	0.202

Overall, the results reveal a wide performance plateau across all considered hyperparameters, indicating that the gains achieved by the Two-Stage approach are driven by its structural decomposition rather than by fragile tuning.

2) Transformer Fine-Tuning: Transformer models are fine-tuned using a standardized and conservative setup, consistent with common practices in document classification. Given the large training set and the use of pre-trained language models, training is limited to 2–4 epochs to ensure convergence while avoiding overfitting and unnecessary computational overhead. Learning rates are selected from widely adopted fine-tuning ranges to guarantee stable optimization.

Batch size is fixed by hardware constraints and is therefore not treated as a tunable parameter. The learning rate is fixed to $2 \cdot 10^{-5}$ for all experiments. To account for stochastic effects, each configuration is evaluated across multiple random seeds (noseed, 42, 1337, 2024).

a) Intra-seed Stability: Stability with respect to random initialization is assessed by measuring prediction agreement across different seeds, since ground-truth labels are unavailable for the evaluation set. High agreement across all architectures indicates stable convergence and limited sensitivity to stochastic optimization [10]. Table IV reports the resulting agreement statistics for each transformer architecture.

TABLE IV: Intra-seed prediction agreement for transformer models.

Model	**Mean Agreement**	**Std**	**#Pairs**
DeBERTa-512	0.8968	0.0011	3
DeBERTa-256	0.9073	0.0028	3
RoBERTa-512	0.9113	0.0029	3
RoBERTa-256	0.9112	0.0008	3

b) Cross-Architecture Hard-Case Analysis: To assess architectural complementarity, prediction overlap is analyzed on a subset of *hard cases*, defined as samples for which at least two models disagree in their predictions. This definition isolates instances characterized by intrinsic semantic ambiguity, removing trivially solvable cases where all models converge [11]. High cross-architecture agreement on these samples suggests that residual errors are driven by inherent ambiguity in the data rather than by hyperparameter choices or model-specific artifacts, consistent with prior observations that hard examples are few, noisy, and difficult to improve systematically [9]. The resulting agreement patterns are summarized in Fig 5 .

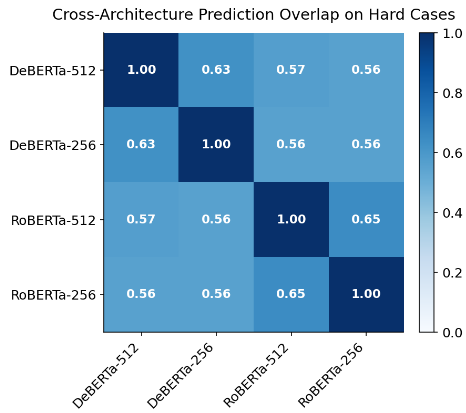

Fig. 5: Cross-architecture prediction agreement on hard cases. Agreement is measured on samples exhibiting prediction disagreement across transformer models, highlighting residual semantic ambiguity between architectures.

Overall, these analyses indicate that transformer fine-tuning operates in a stable regime, with performance differences primarily determined by architectural inductive biases rather than aggressive hyperparameter optimization. Transformer models are therefore treated as robust semantic baselines, whose complementarity is later exploited through ensemble analysis.

III. Results

A. Two-Stage Classification Results

The Two-Stage classifier is evaluated over a wide range of rule-based and linear-model hyperparameters. On the development set, macro-averaged F1 consistently stabilizes around 0.72, indicating a broad near-optimal region rather than sensitivity to a specific configuration.

Bayesian optimization is used to select a representative high-performing setup, whose parameters and corresponding rule coverage are reported in Table V. Comparable performance is achieved by neighboring configurations, confirming robustness to hyperparameter choices.

On the evaluation set, the selected model achieves a macro-F1 of 0.731. The rule-based stage deterministically classifies approximately 16% of samples, while the remaining instances are handled by the probabilistic classifier, highlighting that performance gains stem primarily from the structural two-stage decomposition rather than aggressive tuning.

TABLE V: Selected hyperparameters from Bayesian Optimization and rule coverage for the Two-Stage classifier.

Component	Value
Min rule support	20
Min rule purity	0.984
Rule coverage (Eval)	0.164
Word n-gram max	2
Character n-gram max	5
Min document frequency	3
Max document frequency	0.908
Regularization C	0.842

B. Transformer and Ensemble Results

Transformer models are evaluated across RoBERTa and DeBERTa architectures with input lengths of 256 and 512 tokens using no-seed configurations for fair comparison. As shown in Table VI, all variants achieve tightly clustered macro-F1 scores on the development split, with RoBERTa at maximum sequence length 512 performing best. This limited performance spread suggests diminishing returns from single-model scaling. Multiple random seeds are therefore used mainly to assess optimization stability and prediction agreement, as seed-level performance differences are empirically negligible.

TABLE VI: Best no-seed transformer models on the development split.

Model	Max Len	Best Epoch	Macro-F1 (DEV)
DeBERTa	256	2	0.743
DeBERTa	512	2	0.747
RoBERTa	256	3	0.749
RoBERTa	512	3	**0.756**

Final predictions are obtained via a pruned majority-vote ensemble over multiple transformer instances. Pruning is guided by development-set performance and prediction agreement, retaining only consistently strong and complementary models. Majority voting is adopted as it operates at the decision level and remains robust under distribution shift and class imbalance, where probability-based aggregation may degrade [12]–[14].

This pruned majority-vote ensemble achieves a macro-F1 score of **0.761** on the evaluation set, representing the best-performing configuration submitted.

IV. Discussion

Although the Two-Stage linear model yields only marginal gains in macro-F1, it offers valuable interpretability by isolating near-deterministic cases and highlighting the role of editorial styles and recurring HTML structures, which are characteristic of SEO-oriented publishing practices rather than semantic content alone [15], [16].

Transformer models instead emphasize contextual semantics, surpassing the linear performance plateau but exhibiting higher sensitivity to distributional differences between development and evaluation data, resulting in increased volatility at test time.

Across both regimes, robustness is prioritized over peak performance: decision-level ensembling via pruned majority voting improves stability by aggregating complementary predictors, leading to more reliable generalization under potential distribution shift while preserving strong macro-F1 performance.

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